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Determination of optimal volume of temporary storage tanks in a ship-based carbon capture and storage (CCS) chain using life cycle cost (LCC) including unavailability cost

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ABSTRACT

This study estimated the life-cycle cost (LCC) of a ship-based carbon capture and storage (CCS) chain to determine the optimal volume of temporary storage tanks. The temporary storage tanks should be installed in the ship-based CCS chain to enable ship transportation. Concepts of the ship-based CCS chain were suggested with different volumes of the temporary storage tanks. They were analyzed from an economic viewpoint by LCC analysis. Although general LCC is composed of capital expenditure (CAPEX) and operation expenditure (OPEX), this study additionally considered the unavailability cost, which was derived from the availability. The availability indicates the operational effectiveness depending on the volume of the temporary storage tanks. For estimating the availability, factors causing production loss in the chain were defined first. The failure frequency and restoration time of the defined factors were collected and then the chain was modeled. Finally, the unavailability cost was estimated. CAPEX and OPEX were calculated using a commercial tool and relevant literature. The results showed that the unavailability cost decreased with the increment of the volumes of the temporary storage tanks. In contrast, the CAPEX and OPEX increased with the volumes. When the temporary storage tanks had the same volume of a CO_2 carrier, the lowest LCC was indicated. A sensitivity analysis was performed to investigate the influence of different design parameters and assumptions: CO_2 trade cost, the number of CO_2 carriers, cargo volume of a CO_2 carrier, and distance from a source to a sink.

1. Introduction

Global warming caused by CO_2 emissions is a critical issue in the world (Stocker et al., 2013). To limit the growth of the average global temperature requires massive deployment of various clean energy technologies, including renewable energy, nuclear energy, cleaner transport technologies, energy efficiency, and carbon capture and storage (CCS). CCS offers a critical solution for dealing with CO_2 emissions from fossil fuel use as long as fossil fuel play dominant role in our economic. CCS is only large-scale mitigation option available to reduction in the emissions from industrial sources such as cement, iron and steel, chemicals and refining. CO_2 emission from industrial sources are around 25% of global emission (IEA, 2012; IEA 2013; Global CCS Institute, 2015).

CCS consists of three parts: capture, transportation, and storage. CO_2 is captured at power plants or industrial sites, and the captured CO_2 is injected into onshore or offshore geological storage sites. When an offshore storage site is selected for CO_2 sequestration, CO_2 is transported by offshore pipelines or ships. Pipeline-based transportation is a continuous process, whereas ship-based transportation is a discontinuous process. In order to connect the discontinuous route temporary storage tanks should be installed in ship transportation. It contains the CO_2 produced during the ship voyage. One of the challenges in the design of a ship-based CCS chain is to determine the optimal volume of the temporary storage tanks.

Several volumes of the temporary storage tanks were proposed in previous studies. Berger et al. (2004) studied a CO_2 supply chain to apply enhanced oil recovery (EOR) to the Gullfaks field in the North Sea and concluded that an incentive would be needed to achieve acceptable project economics. They assumed that the storage capacity would have to be 150% of the loading capacity of the ship. Aspelund et al. (2006) developed a CO_2 shipping concept and performed cost, energy, exergy, and CO_2 emission analyses. They set the storage capacity to 150% of the ship capacity. Decarre et al. (2010) described a CO_2 transport chain from capture to storage and showed that CO_2 transport by ships was a better option than by pipelines when the distance exceeded 350 km.

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Nomenclature

A(t)	Availability at time t
A_{av}	Average availability during the specified period
A_{year}	Amount of CO_2 to be settled per year
ASTM	American Society for Testing and Materials
BOG	Boil-off gas
CAPEX	Capital expenditure
CDF	Cumulative distribution function
LCC	Life-cycle cost
$N_{V,year}$	Number of voyages per year
OPEX	Operation expenditure
RBD	Reliability block diagram
V_{SHIP}	Cargo volume of CO_2 carrier
V_{ST}	Volume of temporary storage tanks

They determined the capacity of temporary storage to be 200% of that of the ship for a safety margin. The European Technology Platform for Zero Emission Fossil Fuel Power Plants (ZEP, 2011) analyzed CO_2 transport costs, which included onshore and offshore pipelines and ship transport (including liquefaction). They concluded that combining pipelines and ships could provide cost-effective solutions. They assumed the capacity of a hub terminal to be 150% the size of a ship. Vermeulen (2011) developed a concept of CO_2 liquid shipping system and proposed a complete logistical solution. They set the size of a hub terminal to 150% the ship size. Lee et al. (2011) modeled an intermediate CO_2 storage system for long-distance shipping and calculated the size of a storage tank using the input flowrate. The capacity of the storage tank was assumed to be 133% the cargo capacity of a CO_2 carrier. Yoo et al. (2013) evaluated the feasibility of CO_2 transport by ship for a commercial project and showed that a CO_2 terminal and large CO_2 carriers reduced transport costs. They considered 120% of the ship capacity for the size of the temporary storage tank because it provided a greater operational margin than 100% of the ship capacity and more cost-effectiveness than 150% of that. Roussanaly et al. (2013) illustrated a CCS chain methodology and the functionality of the transport assessment modules from the BIGCCS Research Centre. They considered the volume of buffer storage to be equal to the ship's cargo volume. Table 1 tabulates the suggested volumes of temporary storage tanks in previous studies. Although many studies suggested various cargo volumes for the temporary storage tanks, from 100% to 200% of the cargo volume of the ships, no studies conducted an analysis of the volume of the temporary storage tanks from the viewpoint of an operational margin in our best knowledge.

The operational margin defines by how much a system is designed to be higher than the requirement for normal operation to prepare for unexpected events. If a ship becomes inoperative due to some problems under a small volume of temporary storage tanks (low operational margin), the whole chain should be stopped, as the small volume cannot afford to buffer the problems. In contrast, it is unnecessary to stop the whole chain when a large volume of temporary storage tanks (high operational margin) is installed.

The operational margin can be evaluated by availability. Availability shows the extent to which a system approaches stable operation without production loss. Availability assessments are frequently performed in the oil and gas, chemical, and power plant industries to find optimum design options, to predict production levels, to achieve regularity for financial planning, and to evaluate maintenance and operating policies.

Many studies employed the concept of availability for evaluating systems. Bosman (1985) analyzed the availability of a natural gas compressor plant. Taylor et al. (2000) developed a model for the analysis of a fusion power plant. Hajeeh and Chaudhuri (2000) estimated the availability of a reverse osmosis plant that produces potable water from seawater. Marseguerra and Zio (2000) proposed a new systematic procedure and estimated the production availability of an offshore installation. Crespo Marquez et al. (2005) proposed a general approach using a Monte Carlo simulation to assess the availability and reliability of a complex system and applied it to two configurations of a cogeneration plant. Chang et al. (2008) compared the availability of conventional and new propulsion systems for LNG carriers. Although many studies estimated the availability, few studies worked out the availability including a transportation system that includes trucks, trains and ships. The availability including the transportation system is estimated in this study because a ship-based CCS chain contains a CO_2 carrier.

The optimal volume of the temporary storage tanks should be determined by considering not only the availability but also the costs. A larger volume of temporary storage tanks provides stable operation by allowing delays of ship transportation, but it is accompanied by a high cost. A smaller volume requires a lower cost, but it risks unstable operation. The life-cycle cost (LCC) is employed to evaluate the volume of the storage tanks. Although a typical LCC consists of capital expenditure (CAPEX) and operation expenditure (OPEX), this study additionally considers the availability, which is converted into the unavailability cost to consider it together with the CAPEX and OPEX. The LCC in this study includes the CAPEX, OPEX, and unavailability cost.

The purpose of this study is to determine the optimal volume of the temporary storage tanks in a ship-based CCS chain using LCC including the availability. The structure of this study is as follows. A ship-based CCS chain is described in Section 2, including the suggested design variables. In Section 3, the approaches, including the methodologies for the availability and cost estimation, are discussed. The results are shown in Section 4. Finally, the conclusions are presented.

2. Chain description

2.1. Target source and sink

The Bo-ryeong coal-fired electrical power plant is selected for the assessment of CO_2 capture, and the Ul-leung basin is nominated as the sequestration site, as shown in Fig. 1. Ul-san is determined as the intermediate terminal. The distance from the Bo-ryeong coal-fired power plant to the Ul-san intermediate terminal is approximately 720 km by ship, and the disposal amount of CO_2 is assumed to be 1 million tons per year. This is the target amount of the Korea CCS demonstration project. The life span of the chain is assumed to be 20 years.

2.2. Design variables

This study proposes a concept of a ship-based CCS chain, as shown in Fig. 2 (Seo et al., 2016). The chain consists of a capture system, a liquefaction system, temporary storage tanks, a cargo-handling system, a CO_2 carrier, storage tanks in an intermediate terminal, and a pumping system. First, CO_2 from the power plant is captured by the capture

Table 1
Suggested volumes of temporary storage tanks in previous studies.

References	% cargo volume of ships
Berger et al. (2004)	150
Aspelund et al. (2006)	150
Decarre et al. (2010)	200
ZEP (2011)	100
Vermeulen (2011)	150
Lee et al. (2011)	133
Yoo et al. (2013)	120
Roussanaly et al. (2013)	100

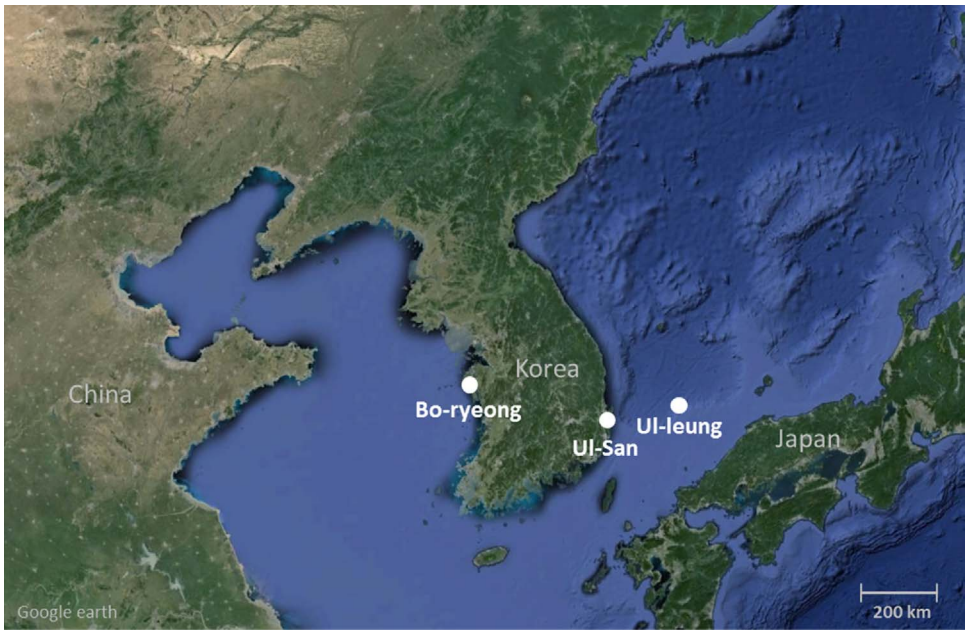


Fig. 1. Target source and sink for a Korean ship-based CCS chain.

system and liquefied for economic transportation. The liquefied CO_2 is stored in the temporary storage tanks and unloaded to the CO_2 carrier by the cargo-handling system. The CO_2 carrier transports the liquefied CO_2 to an intermediate terminal, and then the CO_2 is pressurized by the pumping system to be transported by pipeline to a storage site.

The dashed boxes in Fig. 2 show the design variable analyzed in this study, the volume of the temporary storage tanks, V_{ST}. Eleven values range from 100 to 200 in increments of 10 are suggested for V_{ST} and analyzed to determine the optimum. $V_{ST,100}$ means that V_{ST} is identical to the cargo volume of the CO_2 carrier, V_{SHIP}, and $V_{ST,200}$ indicates that V_{ST} is 200% of V_{SHIP}. This study does not investigate the storage tanks in the intermediate terminal because the operational margin is scarcely affected by its volume. This study assumes that there is no malfunction caused by unpredicted well (or reservoir) conditions.

V_{SHIP} should be determined first because it provides the basis of V_{ST}. V_{SHIP} is calculated by the number of voyages per year, $N_{V,year}$, divided into the amount of CO_2 to be settled per year, A_{year}, as shown in Eq. (1). When the unit of A_{year} is kg, it should be divided by its density, ρ (kg/m^3). $N_{V,year}$ is estimated using the voyage time per a round trip, as shown in Eq. (2). The voyage time is sum of the loading/unloading time, harbor in/out time and transportation time. V_{SHIP} is divided by 0.98 to take the filling limit into account. The International Code of the Construction and Equipment of Ships Carrying Liquefied Gases in Bulk (IGC Code) recommends that the maximum liquid volume in a cargo tank is less than 98% for type-C tank application to avoid tank explosion by liquid expansion (IMO, 2003). When A_{year} is 1 millon tons per year (10^9 kg/year) and the voyage time is 107.8 h, V_{SHIP} is approximately 11,801 m^3. The detail of the voyage time is described in Section 2.3.4.

$$V_{SHIP} = \frac{(A_{year}/\rho)}{N_{V,year}} \times \frac{1}{0.98} \quad (1)$$

$$N_{V,year} = \frac{365 \times 24(hours)}{voyagetime(hours)} \quad (2)$$

The density of the liquefied CO_2 is assumed to be 1064 kg/m^3. In this study, the liquefaction pressure of the CO_2 is assumed to be 15 bar (−27.7 °C) (Seo et al., 2016). Although many previous studies suggested the liquefaction pressure near the triple point due to relatively low pressure and high density, the high refrigeration energy was not properly considered.

2.3. Description of modules

Five modules in the ship-based CCS chain are described: the liquefaction system, the temporary storage tanks, the cargo-handling system, the CO_2 carrier, and the pumping system. The following shows the detail of each system.

2.3.1. Liquefaction system

The liquefaction system converts the captured CO_2 to a liquid due to volume efficiency. Several liquefaction systems are possible for CO_2. In this study, the closed system is employed because it shows the highest efficiency (Seo et al., 2015). The closed system is divided into two parts, as shown in Fig. 3: a compression part and a refrigeration part. The compression part compresses CO_2 because CO_2 is only liquefied above the triple point (5.18 bar, −56.6 °C). After compression, the CO_2 is cooled by the refrigeration part to the liquefaction temperature.

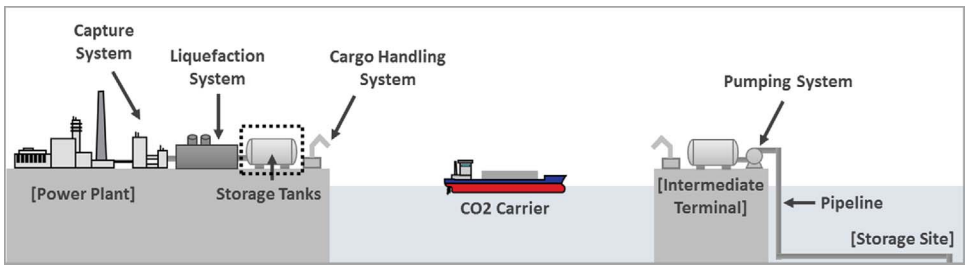


Fig. 2. Concept of ship-based CCS chain (Seo et al., 2016).

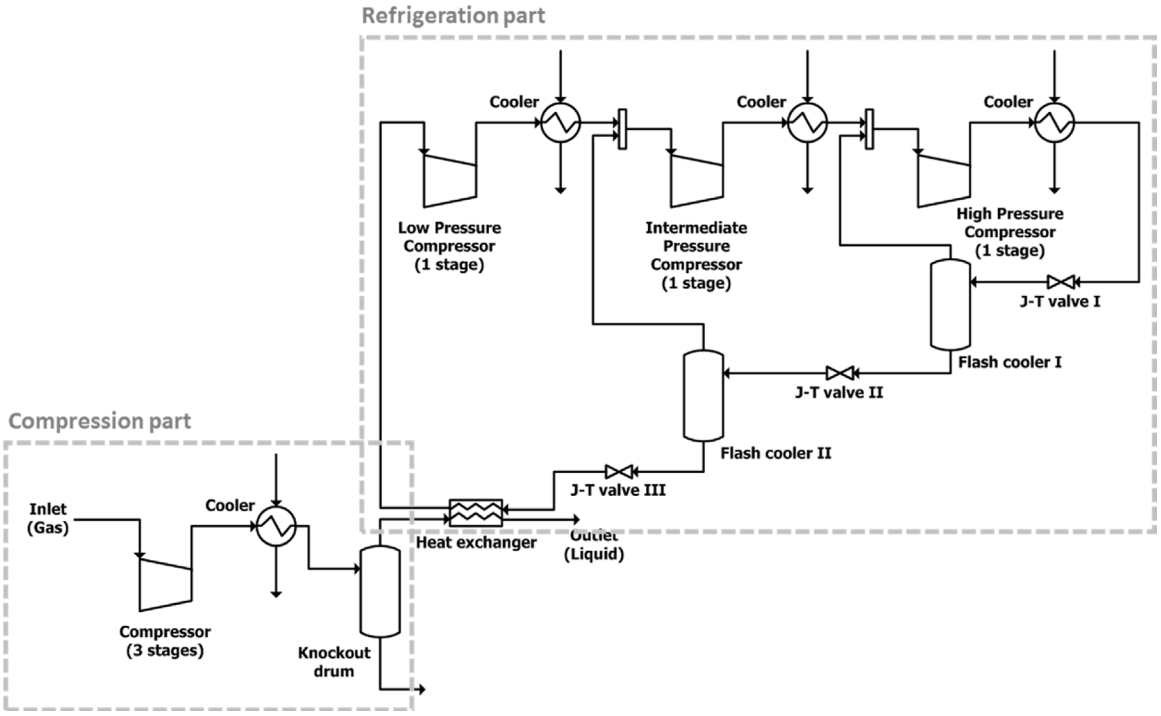


Fig. 3. Closed CO_2 liquefaction system.

2.3.2. Temporary storage tanks

A pressure vessel should be utilized for liquefied CO_2 storage. Cylindrical and spherical vessels are option for the storage of a pressurized fluid. Although the spherical pressure vessel has a lower surface than the cylindrical pressure vessel for the same volume, it is more difficult to manufacture and requires a relatively large space for installation. In this study, the cylindrical pressure vessel is selected for the temporary storage tanks.

2.3.3. Cargo-handling system

The cargo handling systems are installed both on the harbor and on the CO_2 carrier. The cargo-handling system moves the liquefied CO_2 in the temporary storage tanks to the CO_2 carrier and transfers the liquefied CO_2 in the CO_2 carrier to the storage tanks in the intermediate terminal. The former is installed on the harbor (or berth), and the latter is installed on the CO_2 carrier.

The cargo-handling system is generally composed of pumps and loading arms (or hoses). Situationally, it includes a vaporizer and a re-liquefaction plant. The pumps generate the force to lift the liquefied CO_2 in the temporary storage tanks. Generally, a deep well pump and a booster pump are utilized in series. The impeller of the deep well pump is located at the bottom of the cargo tank, and the pump is driven electrically by a motor that is mounted outside the tank. Booster pumps are mounted on deck in an appropriate discharge line, vertically or horizontally. When the CO_2 carrier discharges the liquefied CO_2 to the storage tanks in the intermediate terminal, the cargo of the CO_2 carrier should be filled up with vapor from the intermediate terminal to prevent the pressure shock caused by the rapid pressure drop. When a vapor from the terminal to the CO_2 carrier is unavailable or insufficient, the liquefied CO_2 is fed to the vaporizer to supply a sufficient vapor return. In LPG and ethylene carriers, the re-liquefaction plants liquefy the boil-off gas (BOG) generated during the voyage to maintain the design pressure in the storage tank. In this study, it is assumed that the BOG should not be liquefied in the CO_2 carrier.

2.3.4. CO_2 carrier

The CO_2 carrier transports liquefied CO_2 from the liquefaction site to the intermediate terminal. The cylindrical pressure vessel is employed, analogously to the temporary storage tanks. Although small CO_2 carriers have been used for beverage and food processing in the European market, these carriers are unsuitable for the bulk transport of CO_2 in CCS (Hegerland et al., 2004). IM Skauge SE has 10,000 m^3 ships designed for carrying liquid CO_2, but their cargo is LPG. Research on the development of a large-quantity CO_2 carrier is ongoing. This study proposes a concept design for the CO_2 carrier in cooperation with MTI (a shipbuilding design company) as shown in Fig. 4. It is assumed that a single CO_2 carrier should be used, the average speed should be 15 knots (the average speed of chemical tankers), the loading and unloading operations should take 24 h each, and the times for harbor entry and exit should be 2 h each. This information is from an expert in a ship building company.

2.3.5. Pumping system

The pumping system increases the pressure and temperature of the liquefied CO_2 to meet the required conditions for pipeline transportation. The transported CO_2 should be kept at a high pressure and a high temperature to overcome the reservoir pressure and to prevent hydrate formation and freezing in cases where CO_2 meets water in the reservoir (You et al., 2014). The pumping system consists of a pump and a heat exchanger to increase the pressure and temperature of the liquefied CO_2.

3. Methodology of evaluation

3.1. Life cycle cost (LCC)

LCC analysis is conducted to compare the suggested V_{ST}. The objective of LCC analysis is to quantify the total cost of ownership of a



Fig. 4. Concept design of CO_2 carrier.

product through its full lifecycle. It includes all costs involved in research and development, construction, operation, and maintenance and disposal (Kawauchi and Rausand, 1999). The LCC analysis provides a significant opportunity to select an optimal design.

The CAPEX is money used to purchase, install and commission a capital asset. The OPEX is money used for operation and maintenance, including associated costs such as logistics and spares (ISO, 2000). This study does not consider any discount rate in OPEX and assumes that the OPEX should be an average value over the life cycle.

3.2. CAPEX and OPEX

Only the temporary storage tanks is considered for the cost estimation of the chain, as shown in Fig. 5 because the costs of the other modules are unchanged with the suggested design variable.

The CAPEX of the temporary storage tanks is estimated using the 'Aspen Process Economic Analyzer' (Aspen Technology, 2013). This tool enables front-end consideration of LCC. The following information is entered for the estimation. The material of the vessel is A-517 (American Society for Testing and Materials, ASTM), which is a high-strength quenched and tempered alloy steel intended for use in pressure vessels. It is assumed that the allowance for corrosion should be 3.175 mm, and the weld efficiency should be 85% (Megyesy, 2008). Maximum size of any cylindrical vessel should be less than 5000 m^3 because the maximum size of an existing cylindrical vessel is approximately 6500 m^3 (CESSCO fabrication & engineering LTD). Table 2 presents the result of cost estimation for individual 5000 m^3 vessel. Two 5000 m^3 vessels and one 1801 m^3 vessel are employed for $V_{ST,100}$ (11,801 m^3). The annual contribution to OPEX, primarily maintenance and repair costs, is estimated to be 5% of the CAPEX (Metz, 2005).

3.3. Unavailability cost

For the calculation of the unavailability cost, the availability is estimated first. According to BS4778 (i.e., British standards), the definition of the availability is the ability of an item under the combined aspects of reliability, maintainability and maintenance support to perform its required function at a specified instant or for a specified period. A differentiation of the two definitions can be expressed using the availability, A(t), at time t and the average availability, A_{av}, during the specified period (Rausand and Høyland, 2004). The availability A(t) is defined in Eq. (3), and A_{av}, which represents the availability in this study, is defined in Eq. (4):

$$A(t) = \Pr(\text{The item is functioning at time } t) \quad (3)$$

$$A_{av} = \frac{Uptime}{Uptime + Downtime} \quad (4)$$

Fig. 6 shows a simplified procedure for the estimation of the availability (Calixto, 2012). First, its boundary is determined, and the factors causing production loss are defined. Second, the required data are collected, and the chain is modeled. Finally, the simulation is conducted by a commercial tool. The following shows the details of each step.

3.3.1. Step 1. Chain configuration

Fig. 7 shows the modules of the ship-based CCS chain and the factors causing production loss. This study investigates the eight factors

Table 2
CAPEX of individual 5,000 m^3 vessel.

Items	Material	Manpower
Equipment & Setting	4,838,400	170,663
Piping	72,515	26,760
Civil	107,163	93,857
Structural Steel	90,486	17,006
Instrumentation	68,482	13,900
Electrical	6,226	3,290
Insulation	261,721	236,133
Paint	9,268	11,759
Subtotal	5,454,261	573,368
Total	6,027,629	

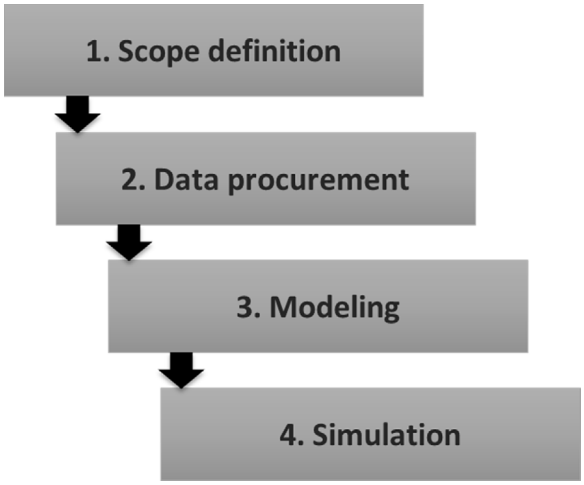


Fig. 6. Procedure for estimation of availability (Calixto, 2012).

that can potentially cause production loss in the chain. In the liquefaction system, the production loss caused by equipment failure and preventive maintenance is considered. This study does not consider any factors for the temporary storage tanks because no factor influences its production loss. In the cargo-handling system, production loss caused by equipment failure is considered analogous to that of the liquefaction system. Equipment failure is considered in the pump system. A reliability block diagram (RBD) is drawn to consider the modules intelligibly. The RBD is a success-oriented logical presentation of the process, and it indicates how a failure affects production loss. Fig. 8 shows the RBD of the liquefaction system.

Four factors that influence the operation of the CO_2 carrier are considered: failure of a ship engine, malfunction of an unloading system, severe weather conditions, and periodic inspection. When the engine of the CO_2 carrier breaks down, the CO_2 carrier is not able to transport CO_2. The engine failure of the CO_2 carrier causes production loss or delay. To consider this impact, the RBD of the engine of the CO_2 carrier is drawn, as shown in Fig. 9. The failure of the unloading system in the CO_2 carrier leads to production loss during the unloading process. Critical weather conditions are considered because the operation of the CO_2 carrier is limited under severe weather conditions. The periodic inspection is considered because the CO_2 carrier is not operated during the inspection. A classification society periodically inspects a ship for

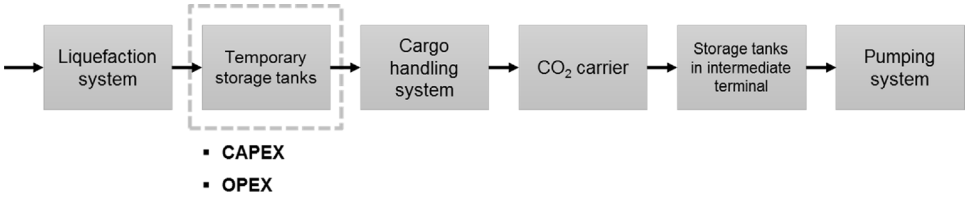


Fig. 5. CAPEX and OPEX consideration of ship-based CCS chain.

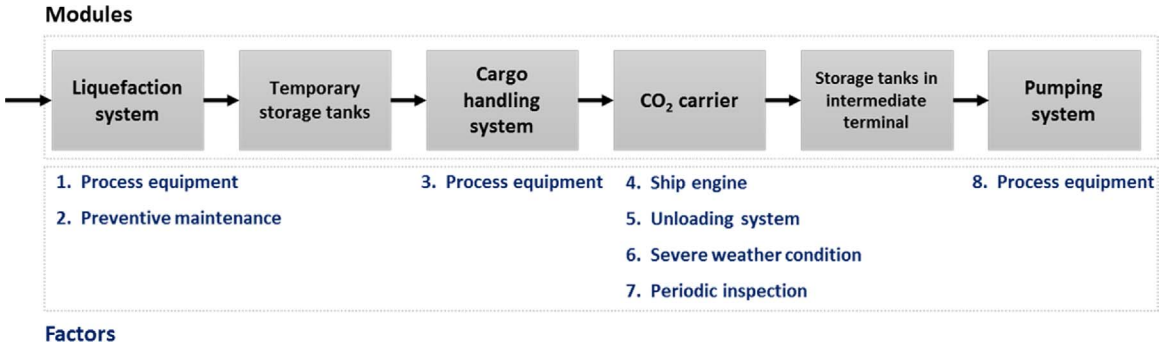


Fig. 7. Configuration of ship-based CCS chain.

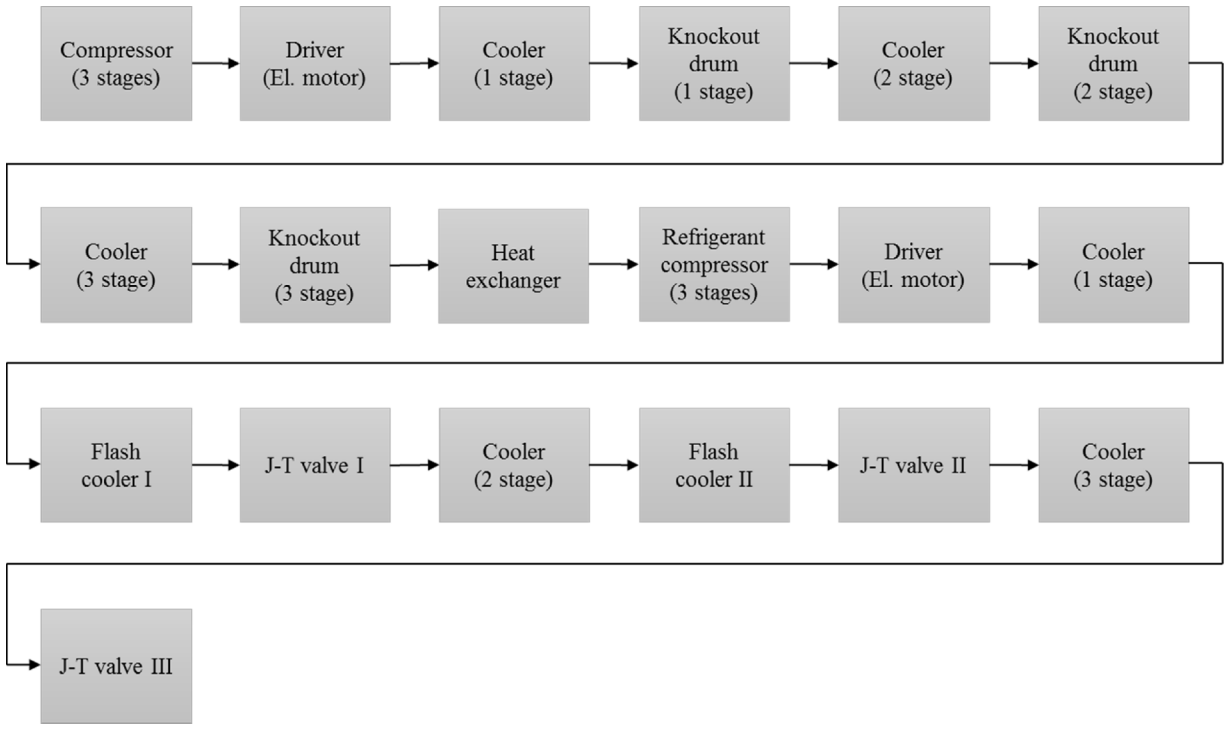


Fig. 8. RBD of liquefaction system.

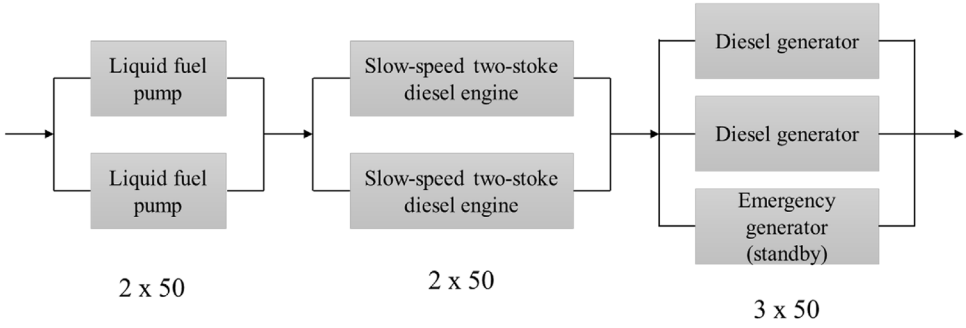


Fig. 9. RBD of engine system of CO_2 carrier.

safety.

3.3.2. Step 2. Data collection

In this step, the reliability and maintenance data are gathered for the estimation of the availability. The reliability data are the frequency of an event, and the maintenance data are the required time for restoration. Those expressions are different depending on the factors. The failure rate and repair time are utilized for process equipment. The frequency and the duration time are employed for preventive maintenance, severe weather conditions and the periodic inspections.

Table 3 shows the failure rate and repair time for all equipment employed in the ship-based CCS chain. These data include the equipment for the liquefaction system, the cargo-handling system, the ship engine, the unloading system in the CO_2 carrier, and the pumping system (SINTEF, 2009). Table 4 shows the information on preventive maintenance collected from the open literature (Juan, 2010).

This study considers a typhoon for the severe weather condition. The operation of the CO_2 carrier is limited when the typhoon passes Korea. Figs. 10 and 11 show the cumulative distribution functions (CDFs) of the typhoon frequency and affecting time. The CDF of the

Table 3
Failure rate and repair time of equipment in ship-based CCS chain.

Component	Mean failure rate (per 10^6 h)	Mean active repair time (h)
Process equipment in liquefaction system		
Compressor (centrifugal)	182.8	49.0
Electrical motor (compressor)	124.7	12.0
Heat exchanger (shell and tube)	13.7	1.5
Heat exchanger (plate fin)	20.52	4.2
Vessel (flash drum)	13.91	4.9
Vessel (separator)	35.92	5.1
Process equipment in cargo-handling system		
Pump (gas processing)	184.08	14.0
Ship engine in CO_2 carrier		
Gas turbine	756.8	23.7
Diesel engine	324.7	78.8
Electric generator	48.9	18.0
Oil-processing pump	98.5	14.6
Electric motor	94.5	9.5
Unloading system		
Pump (gas processing)	184.08	14.0
Heater & vaporizer	13.7	1.5
Process equipment in pumping system		
Pump (gas processing)	184.08	14.0
Heater & vaporizer	13.7	1.5
Seawater pump (seawater lift)	74.28	35.0

Table 4
Preventive maintenance interval and downtime (Juan, 2010).

Task type	Period (month)	Downtime (h)
Detergent washing	2	6
Service/cleaning	4	24
Baroscopic inspection	12	72
Overhaul or replacement	60	120
Planned shutdown	48	240

typhoon frequency is generated by the historical data of typhoons that passed over Korea during the last 60 years. The CDF of the typhoon frequency is modeled by a Weibull distribution function because it exhibits the lowest fitting error among the several candidates, including normal and exponential distribution functions. The parameters of the Weibull distribution function in Fig. 10 for the typhoon frequency are 0.438 (λ, scale parameter) and 1.768 (*k*, shape parameter). The CDF value presents the probability of a typhoon occurrence within a specified time. The likelihood of a typhoon occurrence within 0.5 year is approximately 0.72 (72%). A normal distribution function is employed for the typhoon affecting time, and its mean (μ) and standard deviation (σ) assumed to be 60 and 12, respectively.

According to the Korean Register of Shipping, there are three types of periodic inspection for the safety of a ship: a special survey, an intermediate survey, and an annual survey. The special survey is performed every 5 years, and the intermediate survey is conducted every 3 years. When the special survey overlaps with the intermediate survey, only the special survey is performed. When the special or intermediate survey is conducted, the annual survey is omitted.

3.3.3. Step 3. Modeling & Step 4. Simulation

A commercial code (DNV-GL, TARO) is used for the modeling and simulation. There are various techniques for modeling, such as the RBD, Markov model, FTA and Monte Carlo methods. The first three are analytical approaches, whereas the last is a simulation approach. The analytical approaches have limitations for applications to complex systems (Verma et al., 2010). Hence, this study utilizes the commercial code that is based on the Monte Carlo simulation method.

Finally, the unavailability cost, which is a part of the LCC together with the CAPEX and OPEX, is computed on the basis of the availability. This study assumes that the lost production (unavailability) should be filled by carbon credits, such as certified emission reductions (CERs). The carbon credit is a tradable certificate or permit representing the right to emit CO_2. It can be purchased on the CO_2 trade market. If an owner of a CO_2 emission source does not satisfy the target amount of CO_2 due to production loss, the owner should purchase carbon credits on the CO_2 trade market. The availability is changed to the unavailability cost using the CO_2 trade cost. The CO_2 trade cost in the CO_2 trade market is assumed to be 20 USD per ton.

4. Results and discussion

4.1. CAPEX and OPEX

Fig. 12 shows the CAPEX and OPEX of the temporary storage tanks with V_{ST}. The OPEX is the total operation expenditure for the life span (20 years). The CAPEX and OPEX increase proportionally with V_{ST} because the costs increase with the increment of V_{ST}. When V_{ST} is

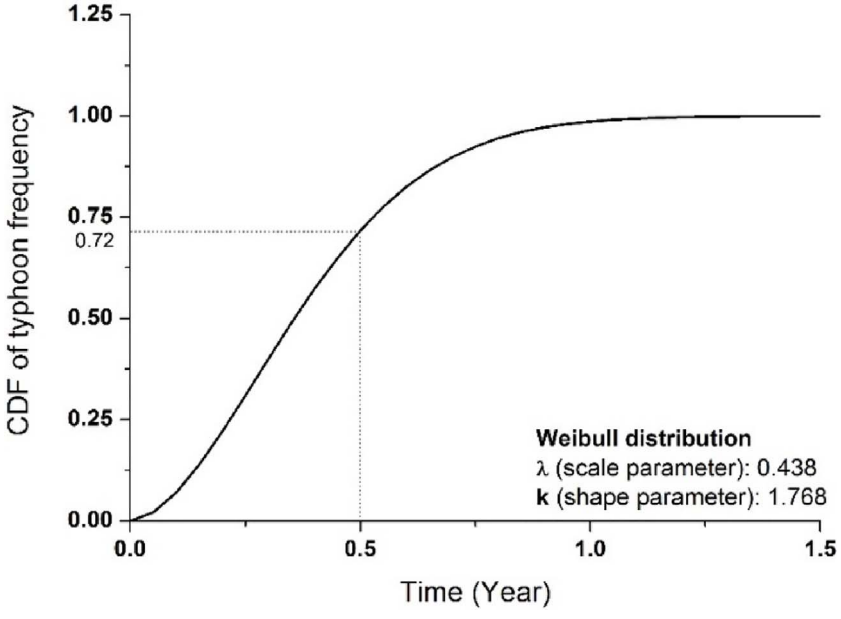


Fig. 10. CDF of typhoon frequency.

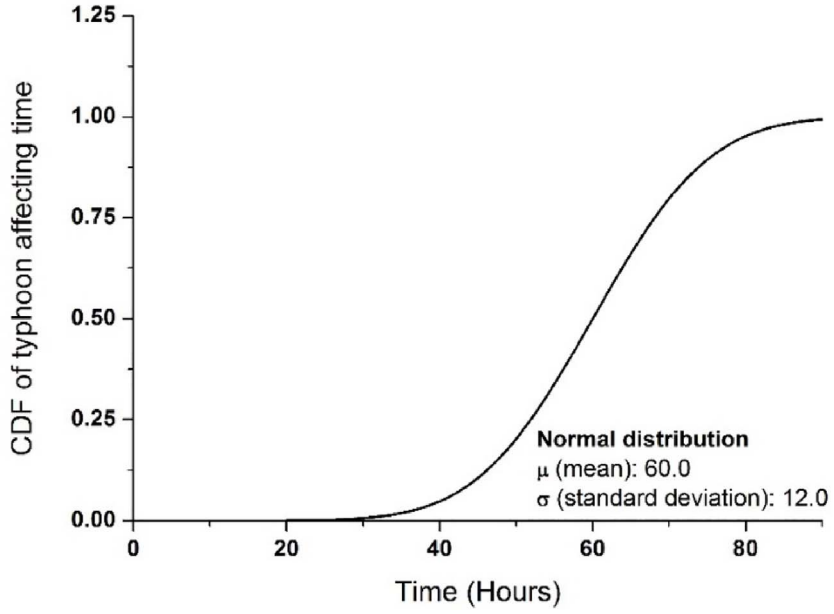


Fig. 11. CDF of typhoon affecting time.

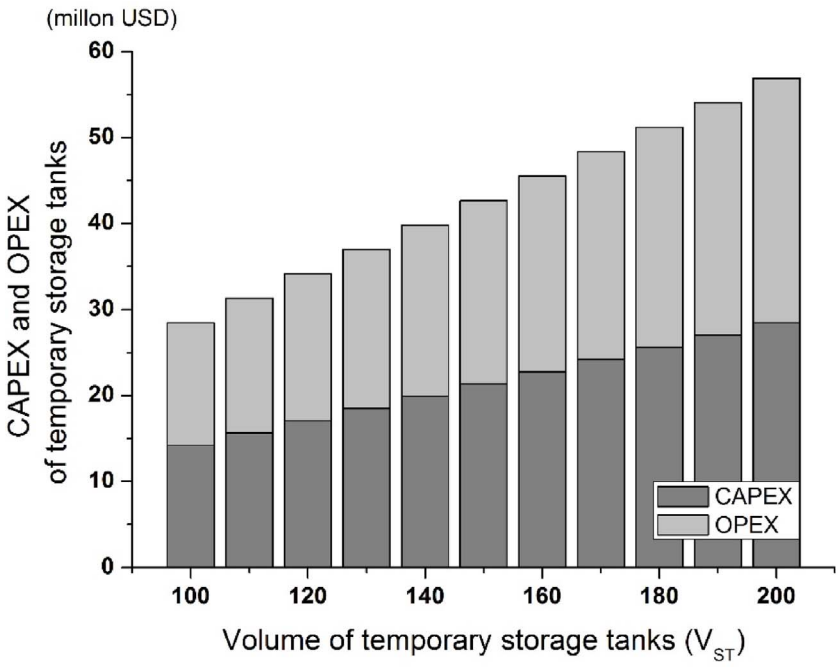


Fig. 12. CAPEX and OPEX with $V_{ST.}$

increased by 10%, additional 2.8 million USD is required.

4.2. Availability and unavailability cost

Fig. 13 indicates the availability and the unavailability cost depending on V_{ST}. The availability increases with V_{ST}, whereas the unavailability cost decreases with it. Since larger V_{ST} allows delays of the CO_2 carrier, the availability grows with V_{ST}. The unavailability cost, which is the required cost to cover the production loss, decreases with V_{ST} because the production loss decreases with the increment of V_{ST}. The availability increases slightly from $V_{ST,150}$ to $V_{ST,200}$ compared to others. This means that the effectiveness decreases as V_{ST} increase. When V_{ST} is increased from 100 to 150, the availability increases 0.86% and the unavailability cost decreases 0.17 million USD.

4.3. LCC

Fig. 14 presents the LCC of the temporary storage tanks with V_{ST}. The LCC is the sum of the CAPEX, OPEX and the unavailability cost. The LCC increases proportionally with V_{ST}. The CAPEX and OPEX contribute ~96% to the LCC, whereas the unavailability cost contributes ~4%. The CAPEX and OPEX are dominant in the LCC. The lowest LCC is indicated at $V_{ST,100}$. This means that the optimal volume of temporary storage tanks is 11,801 m^3, which is identical to the cargo volume of the CO_2 carrier.

4.4. Sensitivity analysis

This study performs a sensitivity analysis to investigate the influences of the given factors (design parameters and assumptions) on the results. It is important to understand the correlation between the factors

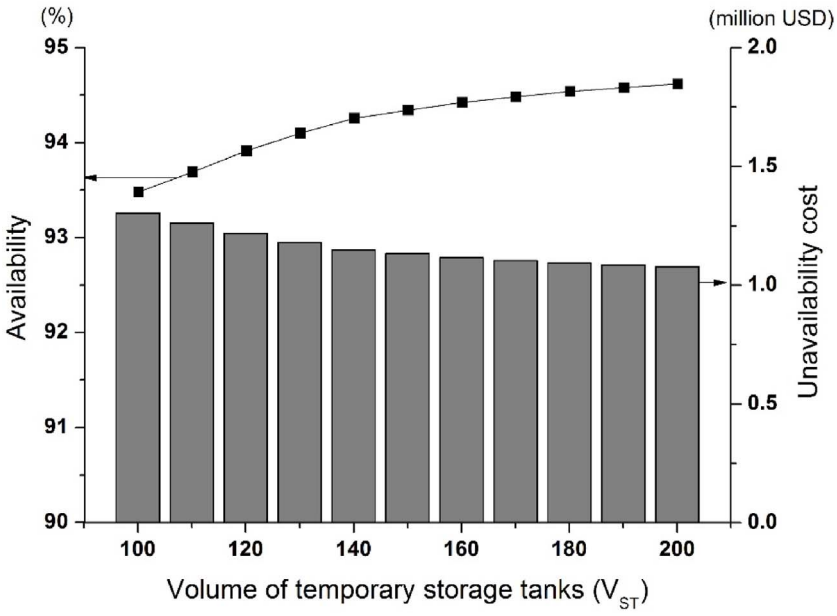


Fig. 13. Availability and unavailability cost with V_{ST}.

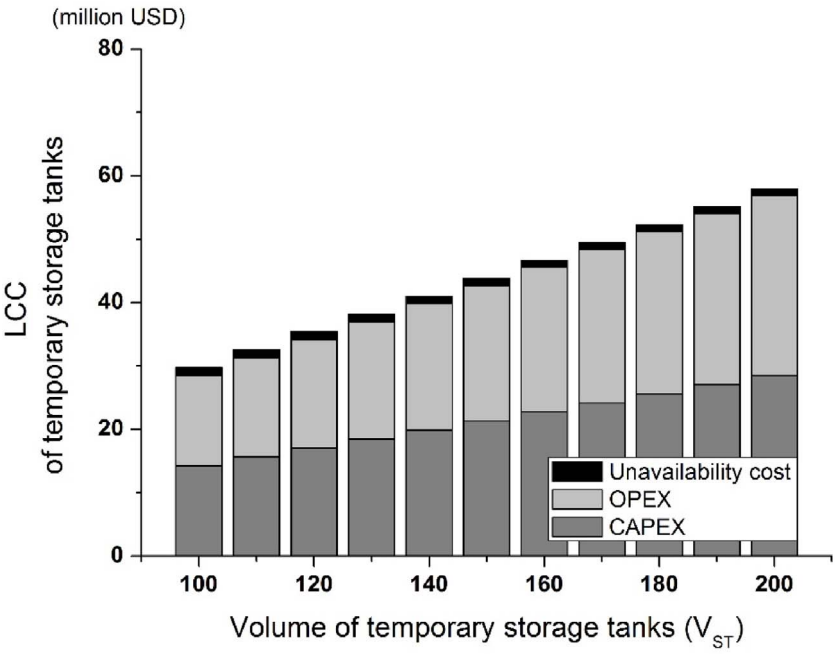


Fig. 14. LCC with V_{ST}.

and the results because the results may be different depending on the given factors. The analyzed factors in this study are different CO_2 trade costs, the different numbers of CO_2 carriers, different cargo volumes of the CO_2 carrier, and the different distance from the source to the sink. The details are presented below.

4.4.1. *Different CO_2 trade cost*

In the previous results the CO_2 trade cost is assumed to be 20 USD per ton to estimate the unavailability cost, but it fluctuates depending on the market status and policies. In addition it considers only the CO_2 trade cost to covert the production loss (unavailability) into the unavailability cost. There can be other loss of benefit. In this study, the LCC of the temporary storage tanks is estimated using five different CO_2 trade costs: 20, 40, 60, 80, 100 USD per ton.

Fig. 15 shows the LCC with different CO_2 trade costs. The LCC increases with the increment of the CO_2 trade cost. The tendency of the LCC does not change with the CO_2 trade cost, indicating that the CO_2 trade cost scarcely affects the results. When the CO_2 trade cost is increased by 20 USD per ton, an additional 1.3 million USD is required. The lowest LCC is changed from $V_{ST,100}$ to others when the CO_2 trade cost reaches 1500 USD per ton. This means that the optimal volume of the temporary storage tanks will be different if the CO_2 trade cost is higher than 1500 USD per ton.

4.4.2. *Different numbers of CO_2 carriers*

The number of CO_2 carriers to transport the CO_2 is fixed as one in the previous results. It can be different depending on the strategy for transportation. Shipping using two CO_2 carriers is analyzed. The cargo volume of each CO_2 carrier in the case of two carriers is half that of the single CO_2 carrier. Since the CAPEX and OPEX of the CO_2 carrier vary with the number of it, both the temporary storage tanks and the CO_2 carrier are considered for the LCC. The following is the description for the CAPEX and OPEX estimation of the CO_2 carrier.

The CAPEX of the CO_2 carrier is divided into two, the costs of the

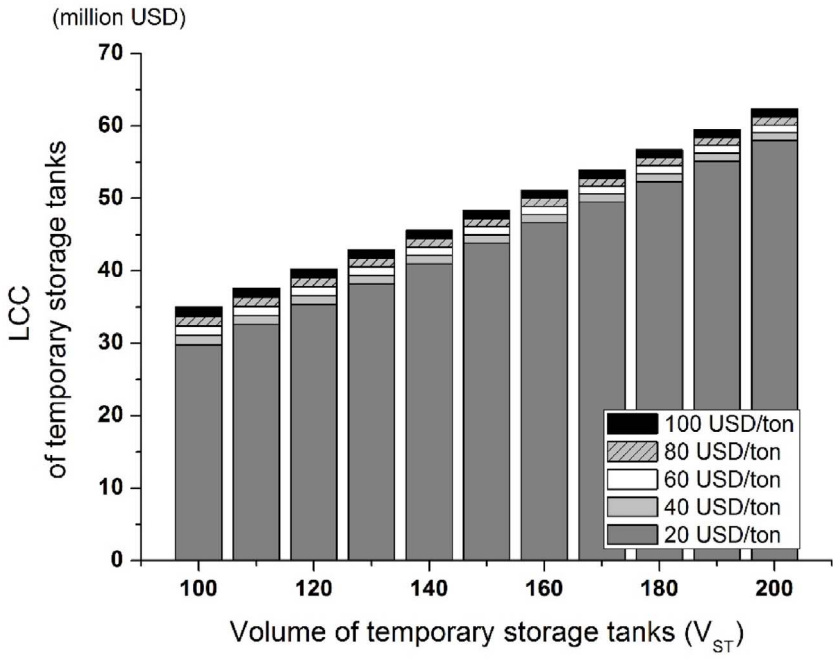


Fig. 15. Chain LCC.

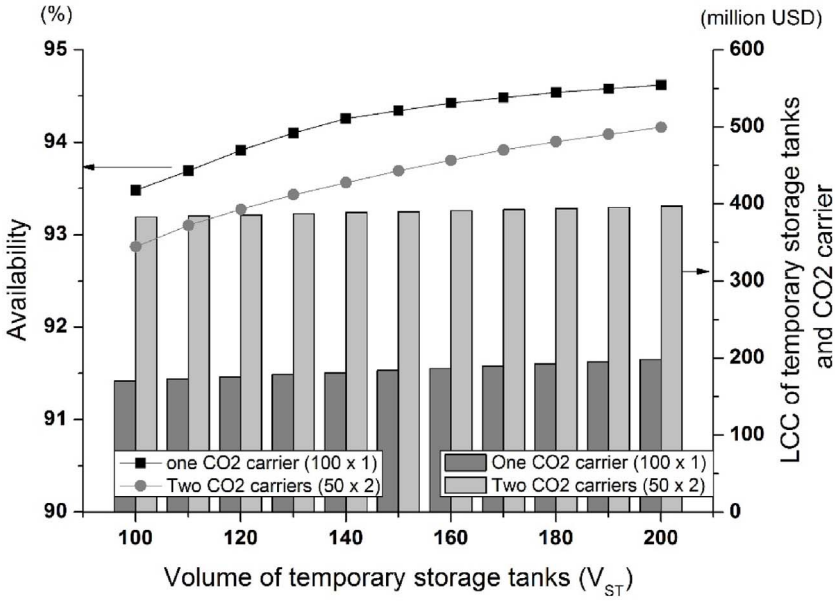


Fig. 16. Availability and LCC with different number of CO_2 carrier.

hull and the cargo tank. The cost of the hull is estimated using that of an oil tanker. Although the hull is slightly different from that of an oil tanker in functions, the main features are similar to each other, and the dominant costs for both are commonly material and manufacturing costs. That is, the cost of the hull is assumed to be that of an oil tanker of the same size (Jung et al., 2013). The assumed crude oil density is 880 kg/m^3 to calculate the volume of the oil tanker. The cost of the oil tanker is estimated using Clarkson's new building prices. The method for the cost estimation for the cargo tank is analogous to that for the temporary storage tank. The annual operating cost of the CO_2 carrier is estimated to be 5% of the CAPEX (Metz, 2005). The fuel consumption cost (voyage cost) is also accounted for in the OPEX. This cost depends on the cruising range and is calculated by the fuel consumption rate of the oil tanker.

Fig. 16 shows the LCC including the temporary storage tanks and the CO_2 carrier and the availability with different number of CO_2 carrier. The availability decreases with the increasing number of CO_2 carriers. The availability of two CO_2 carriers is lower than that of one CO_2 carrier because two carriers experience more events causing production loss than one carrier. The LCC of two CO_2 carriers shows a significantly higher value than that of one carrier. This indicates that a single CO_2 carrier is a better option than two carriers.

4.4.3. *Different volumes of CO_2 carrier*

In the previous results the required cargo volume of the CO_2 carrier is calculated using Eq. (1). It can be larger than the calculated one to provide the operational margin. The CO_2 carrier that has 1.1 times larger cargo volume than the calculated one is analyzed. $V_{SHIP,1.0}$ is the calculated cargo volume of the CO_2 carrier using Eq. (1) whereas $V_{SHIP,1.1}$ is the cargo volume which has 1.1 times larger cargo volume than the calculated one.

Fig. 17 shows the LCC including the temporary storage tanks and the CO_2 carrier and the availability with different cargo volumes. When the enlarged cargo volume of the CO_2 carrier is employed, the

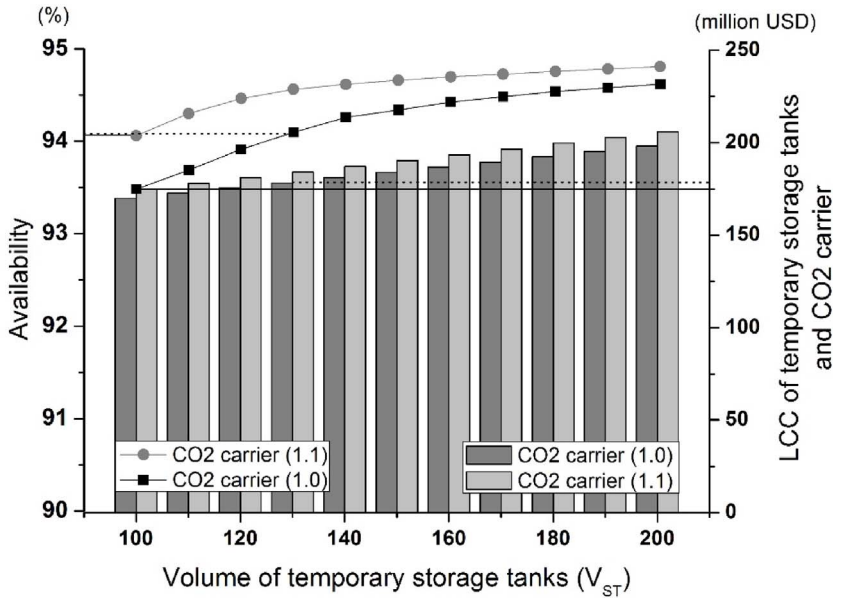


Fig. 17. Availability and LCC with different volumes of CO_2 carrier.

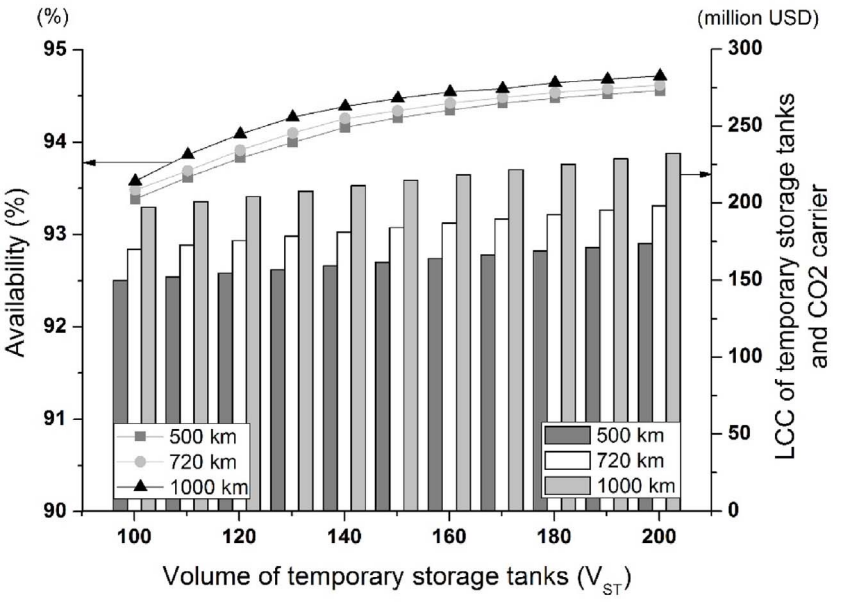


Fig. 18. Availability and LCC with different distance.

availability and the LCC increase. It is apparent that the CO_2 carrier with the larger cargo volume has the higher availability due to its high operational margin. The lowest LCC is indicated when V_{ST} is 100 regardless the volume of the CO_2 carrier. When $V_{ST,100}$ and $V_{SHIP,1.1}$ are used, the availability and the LCC are 94.1% and 175 million USD, respectively. If $V_{ST,130}$ and $V_{SHIP,1.0}$ are employed, the availability and the LCC are 94.1% and 178 million USD, respectively. This result indicates that to increase the cargo volume of the CO_2 carrier is better option than to enlarge the volume of the temporary storage tanks.

4.4.4. Different distance

The target source and sink are the Bo-ryong coal-fired power plant and Ul-leung basin in the previous result. The distance between two sites is 720 km. The target source and sink are changed depending on the national strategy or policies. The availability and LCCs are investigated with three different distances: 500, 720 and 1000 km.

Fig. 18 shows the LCC of the temporary storage tanks and the CO_2 carrier and the availability with different distance. The availability increases slightly with the increment of the distance. The LCC of the temporary storage tanks and the CO_2 carrier increases with the distance because the longer distance requires the larger cargo volume of the CO_2 carrier and the higher operating costs. When the distance is increased from 500 km to 1000 km, the enlarged availability and LCC are 0.2 and 53 million USD, respectively. The lowest LCC is presented at $V_{ST,100}$ regardless the distance.

5. Conclusions

The LCC was estimated with different volumes of the temporary storage tanks to determine the optimum value. The CAPEX and the OPEX were calculated using the commercial tool and the open literature. The availability, which included the transportation system, was estimated, and then it was converted into the unavailability cost to evaluate it together with the CAPEX and the OPEX.

The results showed that the CAPEX and OPEX increased with the volume of the temporary storage tanks, whereas the unavailability cost decreased. The LCC, which is the sum of the CAPEX, OPEX, and the unavailability cost, grew with the volume of the temporary storage

tanks because the CAPEX and OPEX were dominant in the LCC. The CAPEX and OPEX contributed ~96% to the LCC. The lowest LCC was presented when the volume of the temporary storage tanks was identical to that of the CO_2 carrier (11,801 m^3). When the volume of the temporary storage tanks was increased from 100 to 150, the increment of the availability was 0.86 (corresponding to the reduction of 0.17 million USD), but the additional costs were 14.2 million USD.

A sensitivity analysis was performed to investigate the influences of the CO_2 trade cost, the number of CO_2 carriers, the cargo volumes of CO_2 carrier, and the distance. The results of the sensitivity analysis showed that the optimal volume of the temporary storage tanks was unchanged with them. When the CO_2 trade cost approached 1500 USD per ton, the optimal cargo volume of the temporary storage tanks was changed. If a decision maker thought that the improved availability was more valuable than the increased costs, the large volume of the temporary storage tanks could be selected. To increase the cargo volume of the CO_2 carrier is more cost effective solution than to enlarge the volume of the temporary storage tanks for improving the availability (operational margin).

The limitation of this study was that the optimal cargo volume of the temporary storage tanks was only determined on the basis of the economic viewpoint. The unavailability cost was calculated by considering only the carbon credits. The fundamental purpose of CCS was to reduce the CO2 emission from industrial sources for solving the global warming than for taking profits. Other factors like public resistance (social), environmental issues, and associated risks could be more significant than the costs. According to HSE, there are 50% fatalities when the people stay for 10 minutes at the place where the CO2 concentration is approximately 10.5%. The mass CO2 emission due to insufficient storage capacity could be dangerous to employees who work near the storage tanks. The result may be changed depending on the weight of these factors (costs, public resistance, the environmental issues and the related risks). Although this study had some limitation, it provides a meaningful information to the decision maker and to other ship-based CCS projects.

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